

Mixed Membership Estimation in Partial Correlation Network Model*

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Abstract

Latent group structure is a standard device for modeling cross-sectional dependence in large panels, yet existing partial-correlation community methods typically assign each unit to a single group and do not distinguish the formation of conditional-dependence links from their strength. We propose a two-path mixed-membership partial-correlation network (2PMM-PCN). Mixed membership governs link formation along an extensive margin, while node-specific sociability governs link strength along an intensive margin. Although the two paths have distinct economic interpretations, their node-specific components enter the population signal only through a common scale, allowing the mixed memberships to be recovered by a single spectral procedure. We develop PartialCorr-Mixed-SCORE and establish a convergence rate for the average membership estimation error under weak dependence. In an application to U.S. state employment growth, the estimated memberships reproduce the broad group structure found in earlier studies while revealing economically meaningful overlaps that a single-group assignment cannot represent.

Keywords: Concentration matrix; spectral methods; probabilistic graphical model; weighted networks; degree correction

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1 Introduction

Latent group structure is a common way to model heterogeneity and cross-sectional dependence in large panels. Two strands of work dominate. One models grouped heterogeneity in panel regressions, allowing coefficients to vary across latent groups but remain homogeneous within groups ([Bonhomme & Manresa 2015](#), [Su et al. 2016](#), [Zhang et al. 2019](#)). The other recovers groups from the interdependence of large multivariate systems, typically through community or block structures in the implied network ([Guðmundsson & Brownlees 2021](#), [Brownlees et al. 2022](#), [Chen et al. 2023](#), [Zhu et al. 2025](#)). Both strands, however, typically impose pure membership, assigning each unit to a single latent group exclusively. But, in reality, overlapping structure or, equivalently, mixed membership exist everywhere. In addition, in second strand of exploiting network structure, existing literature mainly focus on the mechanism forming potential links in the network, while ignoring explaining the underlying mechanism for weights onto the links. To address both concerns, we instead develop a mixed-membership model for the conditional-dependence structure of a large panel of time series, allowing units to belong to multiple latent groups while separating link formation from link strength.

Two features of existing partial-correlation community models motivate our approach. First, they typically assign each unit to a single group. This is restrictive when units carry diversified exposures: regional economies combine several industries, firms participate in overlapping networks, and financial institutions operate across

business lines. Dependence between two units may therefore reflect more than one latent channel that a single group label cannot record. Second, the same latent structure is typically used to characterize the presence of conditional dependence without separately modeling heterogeneity in the strength of existing links. Yet the determinants of these two margins need not coincide: shared exposures may determine whether two units are conditionally dependent, while size, centrality, or openness may affect the strength of that dependence ([Gabaix 2011](#), [Acemoglu et al. 2012](#)).

We design a model with two paths governing link formation and link strength. The first governs the extensive margin. Each unit carries a membership vector over K latent groups, and a link forms with a probability that depends on these vectors, a group-level connectivity matrix, and a node-specific propensity to connect, as in degree-corrected mixed-membership network models ([Airolidi et al. 2008](#), [Fan et al. 2022](#), [Jin et al. 2024](#)). The second path governs the intensive margin. Conditional on a link being present, its expected weight depends on the sociabilities of the two units. For example, firms in the same industry may be more likely to interact, while the strength of their interaction may vary with how active or complementary they are. Embedding the resulting weighted network in the concentration matrix gives the two-path mixed-membership partial-correlation network (2PMM-PCN), in which the sparsity pattern records which conditional dependencies are present and the normalized link weights determine their strength. Relative to [Brownlees et al.](#)

(2022), memberships may overlap in present paper and existing links may have heterogeneous intensities. Separating link formation from link strength is the main modeling contribution of the paper and parallels the extensive–intensive margin distinction in two-part models for weighted networks (Simpson & Laurienti 2015).

Because the network is latent, the membership structure must be recovered from the concentration matrix estimated from the panel. The node-specific propensity to form links and the node-specific sociability enter the population signal only through their product. This feature preserves the degree-corrected mixed-membership structure required for spectral recovery: an eigen-ratio normalization removes the effective node-specific scale and maps the observations into a simplex whose vertices represent pure membership in the latent groups and whose barycentric coordinates determine the membership weights. This motivates proposed PartialCorr-Mixed-SCORE, which extends SCORE (Jin 2015) and Mixed-SCORE (Jin et al. 2024) from observed networks to partial-correlation networks estimated from panel data.

We establish population identification and sample recovery results. At the population level, the simplex geometry underlying Mixed-SCORE continues to hold under the 2PMM-PCN. At the sample level, the analysis separates two sources of estimation error: randomness in the realized weighted network and estimation of the concentration matrix from the panel. Under weak dependence and an operator-norm covariance-concentration condition, we obtain a convergence rate for the average membership estimation error that makes the contribution of each source explicit.

Monte Carlo simulations show that PartialCorr-Mixed-SCORE recovers mixed memberships well, with accuracy improving as the time dimension, block separation, and share of pure nodes increase. We apply the method to quarterly U.S. state payroll employment growth, studied under single-group assignments by [Hamilton & Owyang \(2012\)](#) and [Brownlees et al. \(2022\)](#). The estimated memberships reproduce their broad three-group structure while revealing economically meaningful overlaps across the latent groups that a hard partition cannot represent.

The paper contributes to the literature on latent group structure in panels by recovering mixed rather than pure memberships from the conditional-dependence structure ([Bonhomme & Manresa 2015](#), [Su et al. 2016](#), [Zhang et al. 2019](#), [Guðmundsson & Brownlees 2021](#), [Brownlees et al. 2022](#)), and to the mixed-membership network literature by separating link formation from link strength and extending spectral recovery to networks estimated from panel data ([Airolidi et al. 2008](#), [Zhang et al. 2020](#), [Jin et al. 2024](#)). It also connects to probabilistic graphical models ([Koller & Friedman 2009](#), [Blei et al. 2003](#), [Munro & Ng 2022](#)) and to panel social interaction structures ([Zhu et al. 2019](#), [Chen et al. 2022](#)).

The remainder of the paper is organized as follows. Section 2 presents the model, estimator, and theory. Sections 3 and 4 report the simulation and empirical results, respectively, and Section 5 concludes.

2 Methodology

This section develops the model and the estimation theory.

2.1 The Two-Path Mixed-Membership Partial-Correlation Network

We model the contemporaneous conditional-dependence structure of an N -dimensional, zero-mean, stationary panel time series $\{Y_t\}_{t=1}^T$ that is Gaussian conditional on the latent weighted network W introduced below. Conditional on W , let $\Sigma = \text{Var}(Y_t | W)$ denote the contemporaneous covariance matrix and let $\Lambda = \Sigma^{-1}$ denote the corresponding concentration, or precision, matrix ([Dempster 1972](#)). All distributional statements concerning Y_t in this subsection are understood conditional on W . The partial correlation between Y_{it} and Y_{jt} given the remaining components satisfies $\rho_{ij} = \text{Corr}(Y_{it}, Y_{jt} | \{Y_{st} : s \neq i, j\}, W) = -\Lambda_{ij} / \sqrt{\Lambda_{ii}\Lambda_{jj}}$ for $i \neq j$. Thus Λ and the partial-correlation matrix share the same off-diagonal sparsity pattern. Under conditional Gaussianity, $\Lambda_{ij} = 0$ is equivalent to conditional independence between Y_{it} and Y_{jt} given the remaining components and W ([Pourahmadi 2013](#)).

Following the normalized-Laplacian construction of [Brownlees et al. \(2022\)](#), we represent this conditional-dependence structure through a latent weighted graph. Let $W \in \mathbb{R}^{N \times N}$ be its symmetric, nonnegative weighted adjacency matrix, with $W_{ii} = 0$. Let $d_{W,i} = \sum_{j=1}^N W_{ij}$ and $D_W = \text{diag}(d_{W,1}, \dots, d_{W,N})$, and use the convention

$(D_W^{-1/2})_{ii} = 0$ whenever $d_{W,i} = 0$. Writing $M_W = D_W^{-1/2} W D_W^{-1/2}$ for the normalized weighted adjacency matrix, the symmetric normalized graph Laplacian is

$$L = I_N - M_W = I_N - D_W^{-1/2} W D_W^{-1/2}. \quad (1)$$

We map the weighted network into the concentration matrix as

$$\Lambda = \frac{1}{\sigma^2} (I_N + \phi L). \quad (2)$$

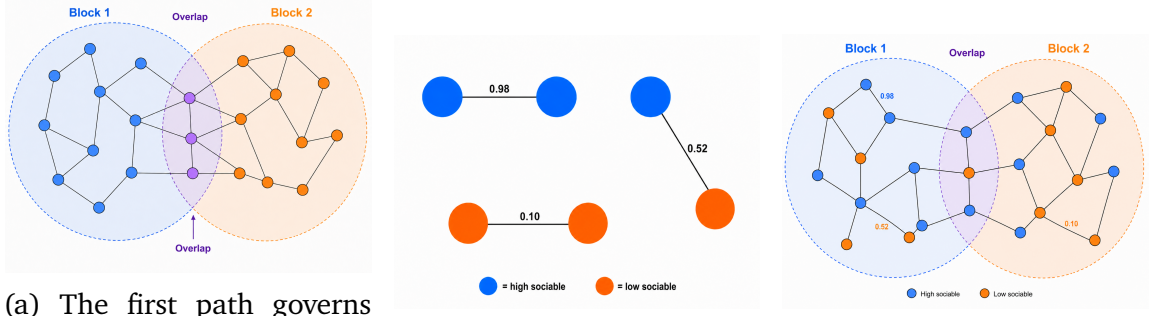
Since L is positive semidefinite, Λ is positive definite for $\phi \geq 0$, and hence defines a valid concentration matrix. Moreover, Λ and L share the same eigenvectors, so the spectral structure used to recover the latent memberships is preserved under this mapping.

Because $W_{ii} = 0$, the diagonal entries of L equal one. Therefore, $\Lambda_{ii} = (1 + \phi)/\sigma^2$. Let $\mathcal{E}_N = \{\min_{1 \leq i \leq N} d_{W,i} > 0\}$ denote the event that the realized weighted graph has no isolated nodes. On $\mathcal{E}_N$, for $i \neq j$, $\Lambda_{ij} = -\phi W_{ij}/(\sigma^2 \sqrt{d_{W,i} d_{W,j}})$ and hence $\rho_{ij} = \{\phi/(1 + \phi)\} W_{ij}/\sqrt{d_{W,i} d_{W,j}} \geq 0$. Under the isolated-node convention above, $\Lambda_{ij} = 0$ whenever either node is isolated. Thus, for $\phi > 0$, $\Lambda_{ij} = 0$ if and only if $W_{ij} = 0$. The sparsity pattern of W therefore determines which pairs of variables are conditionally dependent, while its normalized weights determine the magnitudes of the corresponding partial correlations. When $\phi = 0$, the model reduces to conditional independence.

It remains to specify the latent structure that generates W . Building on [Brownlees et al. \(2022\)](#), we extend their network model in two directions: we replace pure community membership with mixed membership, letting each unit belong to several latent groups, and we add node-specific sociability that governs the intensity of conditional dependence separately from its sparsity pattern. This two-path model generates W , which maps into Λ through (1) and (2). Let $\Pi \in \mathbb{R}^{N \times K}$ collect the mixed-membership vectors $\pi_i \in \mathbb{R}^K$ as its rows, where K is the number of latent groups; Π is our main object of interest, since it determines the latent group structure underlying the partial-correlation network.

The first path governs sparsity. Conditional on (Π, B, Θ) , the upper triangular entries $\{A_{ij} : i < j\}$ are independent Bernoulli variables with success probabilities $\Theta_i \pi_i B \pi_j' \Theta_j$, where $B \in \mathbb{R}^{K \times K}$ is the group-level connectivity matrix, whose diagonal and off-diagonal entries govern within- and between-group connectivity, and Θ is diagonal with entries $\Theta_i > 0$ capturing node-specific propensity to form links. The second path governs link intensity: with $U \in \mathbb{R}^{N \times N}$ diagonal and $u_i \in (0, 1)$ the sociability of unit i , the weight $W_{ij} \mid A_{ij} = 1$ follows a Beta distribution with mean $u_i u_j$ and concentration $\kappa_{ij} > 0$, where larger κ_{ij} tightens dispersion around $u_i u_j$. The two paths thus separate the extensive margin, whether two units connect, from the intensive margin, the expected strength given a connection, while retaining the structure required for spectral recovery.

The two-path mixed-membership network model is defined formally as follows.



(a) The first path governs the sparsity pattern. Mixed memberships generate overlapping latent groups.

(b) The second path governs connection intensity through node-specific sociability.

(c) Together, the two paths determine the weighted adjacency matrix W .

Figure 1: The two-path mixed-membership network construction. The resulting weighted adjacency matrix is mapped into the concentration matrix through (1) and (2).

Definition 1 (Two-Path Mixed-Membership Network Model). *Let $B \in [0, 1]^{K \times K}$ be symmetric and positive definite with $B_{kk} = 1$ for $k = 1, \dots, K$, let $\Theta_i > 0$, let $u_i \in (0, 1)$, and let the rows of Π lie on the simplex, $\pi_i \geq 0$ and $\pi_i \mathbf{1} = 1$. Assume $0 \leq \Theta_i \pi_i B \pi_j' \Theta_j \leq 1$ for all $i \neq j$ and $0 < \underline{\kappa} \leq \kappa_{ij} \leq \bar{\kappa} < \infty$. For $1 \leq i < j \leq N$, the weighted network is generated sequentially by*

$$A_{ij} \mid \pi_i, \pi_j, B, \Theta_i, \Theta_j \sim \text{Bernoulli}(\Theta_i \pi_i B \pi_j' \Theta_j),$$

$$W_{ij} \mid A_{ij}, \kappa_{ij}, u_i, u_j = \begin{cases} \text{Beta}(u_i u_j \kappa_{ij}, (1 - u_i u_j) \kappa_{ij}), & A_{ij} = 1, \\ 0, & A_{ij} = 0. \end{cases}$$

We impose $A_{ji} = A_{ij}$, $W_{ji} = W_{ij}$, $A_{ii} = W_{ii} = 0$, and $\kappa_{ij} = \kappa_{ji}$. Conditional on the latent parameters $(\Pi, B, \Theta, U, \kappa)$, the unordered pairs $\{(A_{ij}, W_{ij}) : 1 \leq i < j \leq N\}$

are mutually independent.

The normalization $B_{kk} = 1$ fixes the group-specific diagonal rescaling inherent in degree-corrected mixed-membership models (Jin et al. 2024). Our identification target is Π . Since U and Θ enter the population signal only through $U\Theta$, they are not separately identified without further restrictions. Under the pure-node condition, however, the rank- K population signal identifies Π up to a permutation of its columns. Thus estimation of Π uses only the first-moment factorization of the weighted network; the two paths are retained to distinguish economically between link formation and link intensity. Formal identification is discussed in Section B.1 of the supplementary material.

The sequential construction in Definition 1, together with conditional independence across unordered pairs, implies

$$\begin{aligned} \mathbb{P}(W, A \mid B, \Theta, \Pi, U, \kappa) &= \prod_{1 \leq i < j \leq N} \mathbb{P}(W_{ij} \mid A_{ij}, \kappa_{ij}, u_i, u_j) \\ &\times \prod_{1 \leq i < j \leq N} \mathbb{P}(A_{ij} \mid B, \pi_i, \pi_j, \Theta_i, \Theta_j). \end{aligned} \quad (3)$$

Under Definition 1, $\mathbb{E}(W_{ij} \mid A_{ij} = 1) = u_i u_j$ and $\text{Var}(W_{ij} \mid A_{ij} = 1) = u_i u_j (1 - u_i u_j) / (1 + \kappa_{ij})$, so κ_{ij} affects dispersion but not the conditional mean. The resulting population signal is given next.

Theorem 1. *Consider the 2PMM of Definition 1 and treat the latent parameters as fixed.*

Define $\Omega_W = U\Theta\Pi B\Pi'\Theta U$. Then, for every $i \neq j$, $\mathbb{E}(W_{ij}) = (\Omega_W)_{ij} = \Theta_i\pi_i B\pi_j'\Theta_j u_i u_j$, whereas $\mathbb{E}(W_{ii}) = 0$.

Thus $\mathbb{E}(W) = \Omega_W - \text{diag}(\Omega_W)$: the two matrices coincide off the diagonal, while $\mathbb{E}(W)$ has a zero diagonal. Here Ω_W is the deterministic low-rank mean structure of W and does not depend on the realized network. The theoretical analysis below retains the full matrix Ω_W , including its diagonal, so as to preserve the exact rank- K factorization (5); the resulting diagonal discrepancy between Ω_W and the realized W is absorbed into the network perturbation of Lemma 3.

Combining the two-path network model with the normalized-Laplacian map yields the partial-correlation network model.

Definition 2 (Two-Path Mixed-Membership Partial-Correlation Network). *Let W be generated according to Definition 1. Let L and Λ be defined by (1) and (2), using the isolated-node convention for $D_W^{-1/2}$ stated above. Conditional on W , the process $\{Y_t\}_{t=1}^T$ is Gaussian with covariance matrix $\text{Var}(Y_t \mid W) = \Sigma = \Lambda^{-1}$.*

Definition 2 formally combines the two layers of the model. The latent parameters $(\Pi, B, \Theta, U, \kappa)$ generate the weighted network W , while W determines the concentration matrix and hence the conditional-dependence structure of Y_t . Under this convention, the symmetric normalized Laplacian is positive semidefinite, so $\Lambda = \sigma^{-2}(I_N + \phi L)$ is positive definite for every $\sigma^2 > 0$ and $\phi \geq 0$. Moreover, under Assumption 2(iv), the expected weighted degree diverges faster than $\log N$,

and a standard concentration argument gives $\mathbb{P}(\mathcal{E}_N) \rightarrow 1$. Hence the isolated-node convention is asymptotically inactive.

The non-negativity of the partial correlations follows from the restriction $W_{ij} \geq 0$. We retain this restriction because [Barigozzi & Brownlees \(2019\)](#) and [Brownlees et al. \(2018\)](#) document that negative partial correlations are rare in network models for economic panels. Allowing signed link weights would require a corresponding modification of the normalization and is outside the scope of the present model.

2.2 Algorithm

We estimate the mixed-membership structure by adapting Mixed-SCORE ([Jin et al. 2024](#)) to the partial-correlation network. Given the panel $\{Y_t\}_{t=1}^T$ and the number of groups $\widehat{K}$ selected by the information criterion introduced above, let $\mathbf{Y} = (Y_1, \dots, Y_T)' \in \mathbb{R}^{T \times N}$ and compute the sample covariance matrix $\widehat{\Sigma} = \mathbf{Y}'\mathbf{Y}/T$. Throughout, we assume $T > N$ and that $\widehat{\Sigma}$ is nonsingular, and set $\widehat{\Lambda} = \widehat{\Sigma}^{-1}$.

Under Definition 2, the sign-reversed off-diagonal part of Λ is a positive scalar multiple of the normalized weighted adjacency matrix $M_W = D_W^{-1/2} W D_W^{-1/2}$ of the realized graph. We therefore define

$$M := -\text{offdiag}(\Lambda). \tag{4}$$

By Definition 2, $M = (\phi/\sigma^2)M_W$, so M , M_W , and L share the same eigenvectors and

the positive factor ϕ/σ^2 does not affect the SCORE ratios. Because M_W already incorporates degree normalization, M retains the degree-corrected mixed-membership structure required by Mixed-SCORE. We therefore apply Mixed-SCORE directly to $\widehat{M} = -\text{offdiag}(\widehat{\Lambda})$, its sample counterpart, without additional degree normalization; the error from replacing Λ by $\widehat{\Lambda}$ is handled by the spectral perturbation analysis below.

Under the conditions imposed below, the leading population eigenvector has strictly positive entries, so the population SCORE ratios are well defined. To stabilize sample ratios when an estimated denominator is small while preserving the rotational formulation used when the non-leading eigenvectors are not individually identified, Step 4 uses Euclidean projection onto a ball. This projection is nonexpansive and orthogonally equivariant. Let $\tau_N > 0$ be a deterministic sequence satisfying $\tau_N \rightarrow 0$. The floor τ_N in Step 7 guarantees that the finite-sample estimator is real valued and is asymptotically inactive because the corresponding population quadratic forms are bounded away from zero after rescaling. The remaining zero-denominator, singularity, and normalization conventions in the algorithm are likewise asymptotically inactive.

Relative to SCORE and Mixed-SCORE, which operate on an observed adjacency matrix, PartialCorr-Mixed-SCORE starts from panel observations, estimates the concentration matrix, and uses its sign-reversed off-diagonal component to recover the normalized latent network; the subsequent SCORE-ratio, vertex-search, and

Algorithm 1 PartialCorr-Mixed-SCORE

Input: Panel data $\{Y_t\}_{t=1}^T$ and estimated number of groups $\widehat{K}$.

- 1: Compute $\widehat{\Sigma} = \mathbf{Y}'\mathbf{Y}/T$ and $\widehat{\Lambda} = \widehat{\Sigma}^{-1}$.
- 2: Form the sign-reversed off-diagonal matrix $\widehat{M}$ whose diagonal entries are zero and whose (i, j) -th entry is $-\widehat{\Lambda}_{ij}$ for $i \neq j$.
- 3: Let $\hat{\lambda}_1$ be the K largest eigenvalue of $\widehat{M}$ and let $\hat{x}_1$ be its corresponding eigenvector. Let $\hat{\lambda}_2, \dots, \hat{\lambda}_{\widehat{K}}$ be the remaining $\widehat{K} - 1$ eigenvalues largest in magnitude, ordered such that $|\hat{\lambda}_2| \geq \dots \geq |\hat{\lambda}_{\widehat{K}}|$, with corresponding eigenvectors $\hat{x}_2, \dots, \hat{x}_{\widehat{K}}$. Choose the sign of $\hat{x}_1$ such that $\mathbf{1}'\hat{x}_1 > 0$, and set $\widehat{X} = [\hat{x}_1, \dots, \hat{x}_{\widehat{K}}]$.
- 4: Set $S = \sqrt{\widehat{K}} \log(N)$. For each i , form the untruncated ratio vector $\tilde{r}_i \in \mathbb{R}^{\widehat{K}-1}$ with entries $\tilde{r}_i(k) = \widehat{X}_{i,k+1}/\widehat{X}_{i,1}$ for $1 \leq k \leq \widehat{K} - 1$, using the convention $\tilde{r}_i = 0$ if $\widehat{X}_{i,1} = 0$. Set $\hat{r}_i = \tilde{r}_i$ if $\|\tilde{r}_i\|_2 \leq S$ and $\hat{r}_i = S\tilde{r}_i/\|\tilde{r}_i\|_2$ otherwise, and let $\widehat{R} = (\hat{r}_1, \dots, \hat{r}_N)'$.
- 5: Apply a vertex-search procedure to the rows of $\widehat{R}$ and obtain the estimated vertices $\{\hat{v}_1, \dots, \hat{v}_{\widehat{K}}\} \subset \mathbb{R}^{\widehat{K}-1}$.
- 6: For each i , solve $(1, \widehat{R}'_i)' = \begin{pmatrix} 1 & \cdots & 1 \\ \hat{v}_1 & \cdots & \hat{v}_{\widehat{K}} \end{pmatrix} \hat{w}_i$; if the augmented vertex matrix is singular, take the minimum-norm solution through its Moore–Penrose inverse.
- 7: For $1 \leq k \leq \widehat{K}$, compute $\hat{b}_{1k} = [\max\{\hat{\lambda}_1 + \hat{v}'_k \text{diag}(\hat{\lambda}_2, \dots, \hat{\lambda}_{\widehat{K}})\hat{v}_k, \tau_N\}]^{-1/2}$.
- 8: For $1 \leq i \leq N$ and $1 \leq k \leq \widehat{K}$, set $\hat{\pi}_{ik}^* = \max\{0, \hat{w}_{ik}/\hat{b}_{1k}\}$.
- 9: If $\|\hat{\pi}_i^*\|_1 > 0$, set $\hat{\pi}_i = \hat{\pi}_i^*/\|\hat{\pi}_i^*\|_1$; otherwise set $\hat{\pi}_i = \widehat{K}^{-1}\mathbf{1}$, for $i = 1, \dots, N$.

Output: Estimated mixed-membership matrix $\widehat{\Pi} \in \mathbb{R}^{N \times \widehat{K}}$.

membership-reconstruction steps follow the Mixed-SCORE construction. We use the Sketched Vertex Search of [Jin et al. \(2024\)](#) in the implementation, while the theory applies to any vertex-search procedure satisfying Assumption 6.

2.3 Theory

This section establishes theoretical guarantees for PartialCorr-Mixed-SCORE. The analysis reflects the two layers of randomness in the model. The first arises from the generation of the weighted network W ; the second arises from observing the panel $\{Y_t\}_{t=1}^T$ conditional on W . We first establish the population simplex geometry associated with the ideal low-rank signal, then control the deviations introduced by the realized network and by estimation of the concentration matrix.

Throughout this section, K may depend on N and T and is allowed to diverge. All constants appearing in the assumptions below are independent of N , T , and K . The results are stated for the true number of groups. If K is estimated, we assume that the selection procedure satisfies $\mathbb{P}(\widehat{K} = K) \rightarrow 1$. Because mixed-membership labels are not intrinsically ordered, estimation error is evaluated up to a permutation of the columns of Π . Model identification is discussed in Section B.1 of the supplementary material.

Define the population expected degrees by $d_{\Omega,i} = \sum_{j \neq i} (\Omega_W)_{ij}$ and let $\mathcal{D}_\Omega = \text{diag}(d_{\Omega,1}, \dots, d_{\Omega,N})$. The population normalized adjacency signal, i.e. the population counterpart of M_W obtained by replacing W and D_W by Ω_W and $\mathcal{D}_\Omega$, is

$$\mathcal{M}_W = \mathcal{D}_\Omega^{-1/2} \Omega_W \mathcal{D}_\Omega^{-1/2} = (\mathcal{D}_\Omega^{-1/2} U \Theta \Pi) B (\Pi' \Theta U \mathcal{D}_\Omega^{-1/2}). \quad (5)$$

We retain the full matrix Ω_W in (5), including its diagonal, because this preserves the rank- K mixed-membership factorization. The realized weighted adjacency matrix W has a zero diagonal; the resulting diagonal discrepancy, together with the sampling variation of W , is treated as part of the network perturbation below.

Assumption 1 (Pure-node representation). *For each latent group $k \in \{1, \dots, K\}$, define $\mathcal{I}_k = \{i \in \{1, \dots, N\} : \pi_i = e'_k\}$, where e_k is the k -th canonical basis vector in $\mathbb{R}^K$. There exists a constant $c_0 > 0$ such that $|\mathcal{I}_k| \geq c_0 N/K$ for $k = 1, \dots, K$.*

Assumption 1 is a separability condition. For fixed K , it reduces to the usual positive-fraction condition. When K increases, each group is required to contain a positive fraction of its natural size N/K in pure nodes. Thus all population simplex vertices remain represented without imposing a fixed positive fraction of the entire cross section for each group.

Assumption 2 (Population regularity). *Let $G = \Pi' \Theta U \mathcal{D}_\Omega^{-1} U \Theta \Pi$. There exist constants $0 < c_B < C_B < \infty$, $0 < c_G < C_G < \infty$, $c_\lambda > 0$, $0 < C_d < \infty$, $0 < c_\phi < C_\phi < \infty$, and $c_\sigma > 0$ such that:*

1. $c_B \leq \lambda_{\min}(B) \leq \lambda_{\max}(B) \leq C_B$ and $c_G \leq \lambda_{\min}(G) \leq \lambda_{\max}(G) \leq C_G$.
2. Ω_W is irreducible.
3. Let $\lambda_1 > \lambda_2 \geq \dots \geq \lambda_K > 0$ denote the K nonzero eigenvalues of $\mathcal{M}_W$. Then $\lambda_1 - \lambda_2 \geq c_\lambda$.

4. With $d_{\min} = \min_i d_{\Omega,i}$ and $d_{\max} = \max_i d_{\Omega,i}$, $d_{\min}/\log N \rightarrow \infty$, $d_{\max}/d_{\min} \leq C_d$,
and $\max_i (\Omega_W)_{ii} \leq C_d$.

5. $c_\phi \leq \phi/\sigma^2 \leq C_\phi$ and $\sigma^2 \geq c_\sigma$.

Assumption 2(i) keeps the rank- K signal uniformly nondegenerate, since the nonzero eigenvalues of $\mathcal{M}_W$ are those of $B^{1/2}GB^{1/2}$ and $\mathcal{D}_\Omega^{-1/2}U\Theta\Pi$ has full column rank. Parts (ii)–(iii) identify the Perron eigenvector while allowing multiplicities among the remaining signal eigenvalues, so the non-leading eigenvectors are identified only as an eigenspace. Part (iv) provides the degree concentration required below while permitting bounded node heterogeneity, and part (v) rules out a vanishing partial-correlation signal.

2.3.1 Population geometry

Let $\mathcal{X} = [x_1, \dots, x_K] \in \mathbb{R}^{N \times K}$ collect an orthonormal basis of the eigenspace associated with the nonzero eigenvalues of $\mathcal{M}_W$, with x_1 denoting the leading eigenvector.

Lemma 1. *Consider the 2PMM of Definition 1 satisfying Assumption 2. For any fixed orthonormal basis $\mathcal{X}$ defined above, there exists a unique nonsingular real matrix $P \in \mathbb{R}^{K \times K}$ such that $\mathcal{X} = (\mathcal{D}_\Omega^{-1/2}U\Theta\Pi)P$.*

For unit i , let $\mathcal{X}_{i,1}$ denote the first entry of the i -th row of $\mathcal{X}$ and let $\mathcal{X}_{i,-1} \in \mathbb{R}^{1 \times (K-1)}$ contain its remaining entries. Define the population SCORE-ratio vector by $r_i = \mathcal{X}'_{i,-1}/\mathcal{X}_{i,1} \in \mathbb{R}^{K-1}$ and let $R = (r_1, \dots, r_N)'$.

Lemma 2. Consider the 2PMM of Definition 1 satisfying Assumptions 1 and 2. Then:

1. The leading eigenvector x_1 can be chosen to have strictly positive entries, so every population SCORE ratio r_i is well defined.
2. There exists a nondegenerate K -vertex simplex $\mathcal{S} \subset \mathbb{R}^{K-1}$ with vertices $v_1, \dots, v_K \in \mathbb{R}^{K-1}$.
3. Every r_i lies in $\mathcal{S}$, and $r_i = v_k$ whenever $\pi_i = e'_k$.
4. Let $w_i \in \mathbb{R}^K$ denote the barycentric coordinate vector satisfying $r_i = \sum_{k=1}^K w_i(k)v_k$. Then $w_i = (\pi'_i \circ b_1) / \|\pi'_i \circ b_1\|_1$, where $\circ$ denotes entry-wise multiplication and $b_{1,k} = [\lambda_1 + v'_k \text{diag}(\lambda_2, \dots, \lambda_K)v_k]^{-1/2}$. Consequently, $\pi'_i = (w_i \oslash b_1) / \|w_i \oslash b_1\|_1$, where $\oslash$ denotes entry-wise division.

Lemma 2 is the population simplex representation underlying Mixed-SCORE. The effective node scale $\mathcal{D}_\Omega^{-1/2}U\Theta$ is diagonal, so the standard SCORE normalization of Jin (2015) removes it. The formulation is invariant to orthogonal rotations of the non-leading signal eigenspace and therefore permits multiplicities among $\lambda_2, \dots, \lambda_K$.

2.3.2 Two-layer spectral perturbation

The estimator acts on $\widehat{M} = -\text{offdiag}(\widehat{\Lambda})$, the sample counterpart of $M = -\text{offdiag}(\Lambda) = (\phi/\sigma^2)M_W$, whereas the population geometry of Lemma 2 is defined

by the ideal signal $\mathcal{M}_W$. The total perturbation can therefore be decomposed as

$$\widehat{M} - \frac{\phi}{\sigma^2} \mathcal{M}_W = (\widehat{M} - M) + \frac{\phi}{\sigma^2} (M_W - \mathcal{M}_W). \quad (6)$$

The first term is the time-series error from estimating the concentration matrix conditional on the realized network; the second is the network error from replacing the ideal normalized signal by the realized weighted network.

Assumption 3 (Time-series regularity and covariance concentration). *Conditional on W , $\{Y_t\}$ is a zero-mean stationary process with covariance matrix $\Sigma = \text{Var}(Y_t | W)$. The standardized process $\{\Sigma^{-1/2}Y_t\}$ satisfies:*

1. *Its conditional absolute-regularity (β -mixing) coefficients satisfy $\beta_Y(m) \leq \exp(-C_1 m^{\gamma_1})$ for constants $C_1, \gamma_1 > 0$.*
2. *For every $x \in \mathbb{R}^N$ with $\|x\|_2 = 1$ and every $s > 0$, $\mathbb{P}(|x' \Sigma^{-1/2} Y_t| > s | W) \leq 2 \exp\{-(s/C_2)^{\gamma_2}\}$ for constants $C_2, \gamma_2 > 0$.*
3. *$1/\gamma = 1/\gamma_1 + 1/\gamma_2$ for some $\gamma < 1$.*
4. *$T > N$ and $N/T \rightarrow 0$.*
5. *Under the joint law of $(W, Y_1, \dots, Y_T)$, $\|\frac{1}{T} \sum_{t=1}^T \Sigma^{-1/2} Y_t Y_t' \Sigma^{-1/2} - I_N\| = O_p(\sqrt{N/T})$.*

Parts (i)–(iii) record primitive absolute-regularity and generalized-exponential-tail conditions of the type underlying covariance concentration for weakly dependent

sequences, in the spirit of Assumption 2 of [Brownlees et al. \(2022\)](#) ([Banna et al. 2016](#), [Chen et al. 2013](#)). The restriction $T > N$, together with the imposed nonsingularity of $\widehat{\Sigma}$, places the analysis in the unregularized regime, while $N/T \rightarrow 0$ is the stronger rate required for operator-norm consistency. Part (v) is the high-level operator-norm concentration condition used directly below, imposed under the joint law of $(W, Y_1, \dots, Y_T)$; no uniform conditional statement over realizations of W is required. We do not claim that parts (i)–(iv) alone imply part (v) without additional restrictions on the joint growth of N and T . The condition holds, for example, in the conditionally i.i.d. Gaussian case when $N/T \rightarrow 0$, and is compatible with stable Gaussian serial dependence under suitable conditions.

Define $\delta_{N,T} = \sqrt{\log N/d_{\min}} + \sqrt{N/T}$.

Lemma 3 (Two-layer matrix deviation). *Consider the 2PMM-PCN of Definition 2 satisfying Assumption 2, with panel $\{Y_t\}$ satisfying Assumption 3. Then*

$$\|M_W - \mathcal{M}_W\| = O_p\left(\sqrt{\frac{\log N}{d_{\min}}}\right), \quad (7)$$

$$\|\widehat{M} - M\| = O_p\left(\sqrt{\frac{N}{T}}\right), \quad (8)$$

$$\left\|\widehat{M} - \frac{\phi}{\sigma^2} \mathcal{M}_W\right\| = O_p(\delta_{N,T}). \quad (9)$$

Equation (7) absorbs both the sampling variation of the weighted edges and the discrepancy between the zero diagonal of W and the diagonal of the low-rank signal

Ω_W . The latter is of order $\max_i(\Omega_W)_{ii}/d_{\min} = O(d_{\min}^{-1})$ and is dominated by the rate in (7). Equation (8) follows from Assumption 3(v), perturbation of the matrix inverse, and $-\text{offdiag}(\Lambda) = M$. The constants are uniform in K under Assumption 2.

Let $\widehat{X}_{-1} = [\hat{x}_2, \dots, \hat{x}_K]$ and $\mathcal{X}_{-1} = [x_2, \dots, x_K]$.

Theorem 2. *Consider the 2PMM-PCN of Definition 2 satisfying Assumption 2, with panel $\{Y_t\}$ satisfying Assumption 3. Let $\hat{\lambda}_1, \dots, \hat{\lambda}_K$ and $\widehat{X} = [\hat{x}_1, \dots, \hat{x}_K]$ be selected as in Algorithm 1. Then $\delta_{N,T} \rightarrow 0$, and with probability tending to one the selected eigenvalues are the K positive signal eigenvalues. There exists a sign $s \in \{-1, 1\}$ such that*

$$\left| \hat{\lambda}_1 - \frac{\phi}{\sigma^2} \lambda_1 \right| + \|s\hat{x}_1 - x_1\|_2 = O_p(\delta_{N,T}). \quad (10)$$

Moreover, there exists an orthogonal matrix $\mathbf{H} \in \mathbb{R}^{(K-1) \times (K-1)}$ such that

$$\|\widehat{X}_{-1}\mathbf{H}' - \mathcal{X}_{-1}\|_F = O_p(\sqrt{K} \delta_{N,T}), \quad (11)$$

$$\left\| \mathbf{H} \text{diag}(\hat{\lambda}_2, \dots, \hat{\lambda}_K) \mathbf{H}' - \frac{\phi}{\sigma^2} \text{diag}(\lambda_2, \dots, \lambda_K) \right\| = O_p(\delta_{N,T}). \quad (12)$$

Theorem 2 treats the leading eigenvector separately from the remaining signal eigenspace. The leading eigenvector is individually identified by the gap in Assumption 2(iii), whereas the remaining $K-1$ eigenvectors are aligned through $\mathbf{H}$ and need not be individually identified. This avoids pairwise eigengaps among $\lambda_2, \dots, \lambda_K$. The

$\sqrt{K}$ factor in (11) is the conversion from an operator-norm subspace perturbation to a Frobenius bound for a $(K - 1)$ -dimensional basis. Under Assumption 4, the sign convention in Step 3 of Algorithm 1 selects the sign s in Theorem 2 with probability tending to one.

2.3.3 SCORE ratios and membership recovery

Assumption 4 (Population delocalization). *There exist constants $0 < c_x < C_x < \infty$ such that $c_x/\sqrt{N} \leq \mathcal{X}_{i,1} \leq C_x/\sqrt{N}$ and $\max_{2 \leq k \leq K} |\mathcal{X}_{i,k}| \leq C_x/\sqrt{N}$ for $i = 1, \dots, N$.*

Assumption 4 is a coordinate-wise delocalization condition. In particular, it permits $\|\mathcal{X}_{i,\cdot}\|_2$ to be of order $\sqrt{K/N}$ and therefore does not implicitly bound K . It also implies that every population SCORE-ratio coordinate is uniformly bounded.

The estimator applies no rotation; $\mathbf{H}$ only aligns the sample and population coordinate systems, and the Euclidean projection commutes with it, so the SCORE perturbation can be analyzed through $\|\mathbf{H}\hat{r}_i - r_i\|_2$.

Theorem 3. *Consider the 2PMM-PCN of Definition 2 satisfying Assumptions 1, 2, and 4, with panel $\{Y_t\}$ satisfying Assumption 3. If $K \log^2 N \delta_{N,T}^2 \rightarrow 0$, then*

$$\begin{aligned} \frac{1}{N} \|\hat{R}\mathbf{H}' - R\|_F^2 &= O_p \left(K \log^2 N \delta_{N,T}^2 \right) \\ &= O_p \left[K \log^2 N \left(\frac{\log N}{d_{\min}} + \frac{N}{T} \right) \right]. \end{aligned} \quad (13)$$

The factor K in Theorem 3 is the dimensional cost of estimating the K -dimensional signal eigenspace. The logarithmic factor arises from the Euclidean truncation radius $S = \sqrt{K} \log N$ used to control rows with small estimated leading coordinates, together with the use of a global Frobenius eigenspace bound. No inverse pairwise-eigengap factor appears because the non-leading eigenvectors are aligned as a single eigenspace.

Let $\mathcal{V} = \begin{pmatrix} 1 & \cdots & 1 \\ v_1 & \cdots & v_K \end{pmatrix} \in \mathbb{R}^{K \times K}$ collect the augmented vertices as its columns.

Assumption 5 (Well-conditioned simplex). *There exist constants $0 < c_{\mathcal{V}} < C_{\mathcal{V}} < \infty$ such that $c_{\mathcal{V}}\sqrt{K} \leq \sigma_{\min}(\mathcal{V})$ and $\|\mathcal{V}\| \leq C_{\mathcal{V}}\sqrt{K}$.*

Assumption 5 is the growing-dimensional analogue of the usual simplex-conditioning requirement. It implies $\|\mathcal{V}^{-1}\| = O(K^{-1/2})$. It also gives the uniform vertex scaling needed below. Indeed, for each k , $\sigma_{\min}(\mathcal{V}) \leq \|\mathcal{V}e_k\|_2 = (1 + \|v_k\|_2^2)^{1/2} \leq \|\mathcal{V}\|$, and hence $\|v_k\|_2 \asymp \sqrt{K}$ uniformly in k . Combining this with Assumption 2(i) and the expression for $b_{1,k}$ in Lemma 2 gives $b_{1,k} \asymp K^{-1/2}$ uniformly in k . Thus these uniform properties are consequences of Assumption 5, rather than additional assumptions.

Let $\mathfrak{P}_K$ denote the set of $K \times K$ permutation matrices.

Assumption 6 (Strongly efficient vertex search). *Let $\hat{v}_1, \dots, \hat{v}_K$ be the vertices returned by the vertex-search step. For the same orthogonal matrix $\mathbf{H}$ as in Theorem 2, there*

exists a random $\mathcal{P} \in \mathfrak{P}_K$ such that, uniformly over the admissible sequence of K ,

$$\max_{1 \leq k \leq K} \|\mathbf{H}\hat{v}_{\mathcal{P}(k)} - v_k\|_2^2 = O_p\left(\frac{1}{N} \|\hat{R}\mathbf{H}' - R\|_F^2\right). \quad (14)$$

Assumption 6 is a high-level geometric stability condition. Its form is the growing- K analogue of the strong-efficiency criterion used for Mixed-SCORE: the same orthogonal transformation that aligns the SCORE coordinates also aligns the estimated simplex vertices. We impose this property directly and uniformly over the admissible sequence of K . Existing strong-efficiency results for SVS are established under additional membership configuration conditions and, in the results used here as motivation, fixed K . Thus the growing- K requirement in Assumption 6 should be viewed as an explicit high-level condition rather than as a consequence of the existing SVS theory. No separate orthogonal-equivariance assumption on the vertex-search map is needed because the required alignment is stated directly in (14).

Combining Theorem 3 and Assumption 6 gives $\max_k \|\mathbf{H}\hat{v}_{\mathcal{P}(k)} - v_k\|_2 = O_p(\sqrt{K} \log N \delta_{N,T})$.

Because the selected eigenvalues estimate $(\phi/\sigma^2)\lambda_k$, the sample scaling vector $\hat{b}_1$ estimates b_1 up to the common factor $(\phi/\sigma^2)^{-1/2}$. The common factor cancels under the final ℓ_1 normalization.

Lemma 4 (Stable scaling vector). *Under the conditions of Theorem 3 and Assumptions 5 and 6, with $\mathcal{P}$ from Assumption 6, $\max_{1 \leq k \leq K} |\hat{b}_{1,\mathcal{P}(k)} / \{(\phi/\sigma^2)^{-1/2} b_{1,k}\} - 1| =$*

$O_p(\log N \delta_{N,T})$. In particular, the quantities inside the square roots defining $\hat{b}_{1,k}$ are strictly positive with probability tending to one, so the floor τ_N in Algorithm 1 is asymptotically inactive.

To see why no additional factor of K appears in Lemma 4, write the sample quadratic form after alignment as $(\mathbf{H}\hat{v}_k)' \{ \mathbf{H} \text{diag}(\hat{\lambda}_2, \dots, \hat{\lambda}_K) \mathbf{H}' \} (\mathbf{H}\hat{v}_k)$. This is exactly equal to the quadratic form used by the algorithm before rotation. By Assumption 5, $\|v_k\|_2^2 \asymp K$ and $b_{1,k}^{-2} \asymp K$. The perturbation of the quadratic form is therefore of order $K \log N \delta_{N,T}$, while the population quadratic form itself is of order K , yielding the relative rate in Lemma 4.

Lemma 5 (Stable barycentric coordinates). *Under the conditions of Theorem 3 and Assumptions 5 and 6, with $\mathcal{P}$ from Assumption 6,*

$$\frac{1}{N} \sum_{i=1}^N \|\mathcal{P}'\hat{w}_i - w_i\|_2^2 = O_p(\log^2 N \delta_{N,T}^2). \quad (15)$$

Lemma 5 follows from $\|\mathcal{V}^{-1}\| = O(K^{-1/2})$. The SCORE-coordinate error in Theorem 3 is therefore reduced by a factor K^{-1} after squaring. The vertex perturbation has the same rate: since w_i is a barycentric coordinate vector, $w_i \geq 0$ and $\|w_i\|_1 = 1$, the perturbation generated by the estimated vertices is a convex combination of the individual vertex errors and is controlled by their maximum rather than by their sum.

The membership reconstruction map combines the barycentric coordinates and the

scaling vector. Since $b_{1,k} \asymp K^{-1/2}$ uniformly in k , the normalization in Lemma 2 implies that the contribution of the barycentric-coordinate error is at most of order K times its average squared ℓ_2 error, while a relative perturbation of b_1 enters without an additional factor of K . Consequently, Lemmas 4 and 5 yield

$$\frac{1}{N} \sum_{i=1}^N \|\mathcal{P}' \hat{\pi}'_i - \pi'_i\|_2^2 = O_p \left(K \log^2 N \delta_{N,T}^2 \right). \quad (16)$$

This gives the main membership result.

Theorem 4. *Consider the 2PMM-PCN of Definition 2 satisfying Assumptions 1, 2, 4, 5, and 6, with panel $\{Y_t\}$ satisfying Assumption 3. If*

$$K \log^2 N \left(\frac{\log N}{d_{\min}} + \frac{N}{T} \right) \rightarrow 0, \quad (17)$$

then

$$\min_{\mathcal{P} \in \mathfrak{P}_K} \frac{1}{N} \|\hat{\Pi} \mathcal{P} - \Pi\|_F^2 = O_p \left[K \log^2 N \left(\frac{\log N}{d_{\min}} + \frac{N}{T} \right) \right]. \quad (18)$$

Theorem 4 establishes consistency of the membership estimator up to a permutation of the latent-group labels while allowing K to diverge, provided its growth is slow relative to the network density and the time-series sample size. The two components of the rate correspond respectively to realization of the weighted network and estimation of the concentration matrix from the panel. The result does not require

pairwise separation among the non-leading signal eigenvalues $\lambda_2, \dots, \lambda_K$, which may be arbitrarily close or have multiplicities; instead it relies on the high-level growing- K vertex-stability condition in Assumption 6. Serial dependence is accommodated through Assumption 3, and no restriction of the form $Y_t = \epsilon_t$ is imposed. The logarithmic loss reflects the use of truncated SCORE ratios together with a global eigenspace perturbation bound.

3 Simulation

We evaluate the finite-sample behavior of PartialCorr-Mixed-SCORE in a Monte Carlo experiment with $K = 3$ communities. To respect the normalization $B_{kk} = 1$, we set the diagonal entries of B equal to one and its common off-diagonal entry equal to $b_{\text{out}}/b_{\text{in}}$. We study three designs indexed by $(b_{\text{in}}, b_{\text{out}}) \in \{(0.25, 0.01), (0.50, 0.01), (0.25, 0.05)\}$, with within-to-between ratios of 25, 50, and 5.

The panel has $N = 120$ series, of which N_0 are pure (affiliated with a single community) and the remaining $N - N_0$ are mixed; we let $N_0 \in \{30, 50, 80, 100, 120\}$. Each pure series takes a canonical membership vector $\pi_i \in \{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$, and the mixed series are split evenly between $\pi_i = [0.5, 0.5, 0]$ and $\pi_i = [0, 0.5, 0.5]$. The pure series are allocated as evenly as possible across the three groups. The socialities u_i are i.i.d. Uniform $[0.3, 1.0]$, and the degree-heterogeneity parameters Θ_i

are equally spaced on $[0.9\sqrt{b_{\text{in}}}, \sqrt{b_{\text{in}}}]$. This parameterization gives exactly the same link-formation probabilities as using diagonal entries b_{in} , off-diagonal entries b_{out} , and degree-heterogeneity parameters equally spaced on $[0.9, 1.0]$, while satisfying the normalization $B_{kk} = 1$ imposed in Definition 1. We set $\kappa_{ij} = 1$ for all $i \neq j$ and set $\phi = \sigma^2 = 1$.

We draw the weighted network from the 2PMM of Definition 1, map it to the concentration matrix Λ through Definition 2, and set $\Sigma = \Lambda^{-1}$. The panel is generated as $Y_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma)$, $t = 1, \dots, T$; the algorithm forms $\hat{\Sigma} = \mathbf{Y}'\mathbf{Y}/T$ and $\hat{\Lambda} = \hat{\Sigma}^{-1}$ and estimates each π_i . Over 500 replications, we report the minimum MSE over all permutations of the group labels, $\min_{\mathcal{P} \in \mathfrak{P}_K} N^{-1} \|\hat{\Pi}\mathcal{P} - \Pi\|_F^2$.

Table 1 collects the results, and PartialCorr-Mixed-SCORE recovers the mixed memberships accurately across the designs. The error declines as T grows, monotonically along every row, which matches the N/T term in the rate of Theorem 4. Accuracy also improves with the strength of the block structure: recovery is most precise under $(0.50, 0.01)$ and least precise under $(0.25, 0.05)$, so at $N_0 = 120$ and $T = 1000$ the errors are 0.013 and 0.108, with $(0.25, 0.01)$ between them at 0.040. A larger share of pure nodes works in the same direction, since extra pure series sharpen the simplex vertices required by Assumption 1: under $(0.50, 0.01)$ at $T = 1000$ the error falls from 0.100 at $N_0 = 30$ to 0.013 at $N_0 = 120$. Graphical counterparts are deferred to Section B.2 of the supplement.

Table 1: Estimation errors of PartialCorr-Mixed-SCORE.

$N = 120$					
N_0	$b_{\text{in}}/b_{\text{out}}$	$T = 300$	$T = 500$	$T = 800$	$T = 1000$
30	0.25/0.01	0.168	0.158	0.150	0.145
	0.50/0.01	0.150	0.122	0.104	0.100
	0.25/0.05	0.172	0.169	0.164	0.162
50	0.25/0.01	0.187	0.170	0.154	0.142
	0.50/0.01	0.151	0.102	0.068	0.056
	0.25/0.05	0.195	0.189	0.182	0.177
80	0.25/0.01	0.195	0.154	0.103	0.090
	0.50/0.01	0.115	0.045	0.026	0.022
	0.25/0.05	0.221	0.203	0.181	0.170
100	0.25/0.01	0.200	0.124	0.069	0.056
	0.50/0.01	0.082	0.031	0.019	0.016
	0.25/0.05	0.235	0.202	0.165	0.147
120	0.25/0.01	0.186	0.089	0.048	0.040
	0.50/0.01	0.057	0.024	0.016	0.013
	0.25/0.05	0.242	0.188	0.128	0.108

Notes: Each entry reports the average, over 500 simulation replications, of the minimum MSE over all permutations of the group labels.

4 Application: Learning the Mixed Memberships of U.S. States

We apply PartialCorr-Mixed-SCORE to state-level payroll employment in the United States. The series are seasonally adjusted, annualized quarter-to-quarter growth rates in payroll employment from 1956 to 2007, taken from the U.S. Bureau of Labor Statistics. Following [Hamilton & Owyang \(2012\)](#) and [Brownlees et al. \(2022\)](#),

we drop Alaska and Hawaii, leaving $N = 48$ states and $T = 207$ quarterly observations. We estimate the latent group structure across states and compare it with the groupings reported in those two studies.

Both studies use this panel but assign each state to a single group. [Hamilton & Owyang \(2012\)](#) specify a Markov-switching model in which states cluster by the timing of their business cycles and estimate the assignments by Markov chain Monte Carlo; cross-validation selects three groups. [Brownlees et al. \(2022\)](#) model the partial-correlation matrix as a stochastic block model and recover hard memberships by spectral clustering. We keep their panel and their three-group choice ($K = 3$) and relax the single-group restriction, letting a state’s employment cycle load on more than one latent group. Relaxing that restriction matters here: a state whose labor market responds to both energy prices and national manufacturing demand has no natural place in a hard partition, and forcing it into one discards information about the composition of its economy.

Table 2 reports estimated memberships for a set of representative states; the full set of 48 appears in Section B.3 of the supplement. Using near-pure states as anchors, we label the three groups as financial and services, oil-producing and agricultural, and manufacturing, which correspond to the financial, oil, and manufacturing groups of [Hamilton & Owyang \(2012\)](#). The estimates reproduce that classification, and the mixed weights add economic content that a hard assignment cannot convey.

Several states with large service-oriented economies place their largest weight on

Table 2: Mixed-membership estimates for selected U.S. states.

State	Financial and Services	Oil-Producing and Agricultural	Manufacturing
Massachusetts	0.736	0.000	0.264
New Jersey	0.777	0.000	0.223
New York	0.524	0.000	0.476
California	0.848	0.032	0.120
Texas	0.062	0.600	0.339
Florida	0.112	0.732	0.155
Missouri	0.129	0.871	0.000
Maine	0.046	0.954	0.000
Louisiana	0.247	0.480	0.273
Michigan	0.602	0.000	0.398
Indiana	0.340	0.000	0.660

the first group: New Jersey (0.777), Massachusetts (0.736), New York (0.524), and California (0.848). The second group collects states whose employment dynamics are more closely associated with energy and agriculture, including Texas (0.600) and Missouri (0.871). Louisiana divides its weight across all three groups (0.247, 0.480, 0.273), consistent with employment dynamics that share features of the energy, manufacturing, and service-oriented groups rather than fitting naturally into a single group.

The mixed weights are particularly informative for states whose employment dynamics are associated with more than one latent group. Indiana remains primarily manufacturing-oriented (0.660) while also carrying a sizable financial-and-services weight (0.340). Michigan loads on both financial and services (0.602) and manufacturing (0.398), while New York displays a similarly mixed profile, with weights 0.524 and 0.476, respectively. These estimates indicate that the employment dynamics of

these states share features associated with more than one latent group, a pattern that a hard partition cannot represent.

5 Conclusion

This paper develops the 2PMM-PCN, a partial-correlation network model in which mixed membership governs the formation of conditional-dependence links and node-specific sociability governs their strength. We propose PartialCorr-Mixed-SCORE to recover the mixed-membership structure from the estimated concentration matrix of a panel. The population analysis establishes the simplex geometry underlying identification, while the sample analysis provides convergence rates for the average membership estimation error under weakly dependent data. The resulting rate separates the variation generated by the realized weighted network from the error arising from estimation of the concentration matrix.

The simulations show that recovery improves with the time-series dimension, stronger separation of the latent groups, and a larger number of pure nodes. In the application to U.S. state-level employment growth, the estimated memberships reproduce the broad group structure found in earlier studies while revealing economically meaningful overlaps that cannot be represented by a hard partition. Mixed membership thus gives an interpretable representation of heterogeneity in partial-correlation networks. Our analysis lets the number of groups grow only within a limited range

and under separation that stays bounded away from zero; relaxing these limits, together with regularized high-dimensional estimation and inference for individual membership vectors, is left to future work.

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